

Business Data Management (Part - 3)

Financial Accounting

1. Meaning of Financial Accounting

Financial accounting is the process of:

**Recording →
Classifying →
Summarizing →
Reporting →
Communicating**

business transactions in financial form.

Its main goal is to **record and report the financial performance of a company** so that users can understand and make decisions.

Accounting = “Language of Business”

Accounting converts day-to-day business activities such as:

- Sales
- Purchases
- Expenses
- Investments

into **structured financial information**.

2. Functions / Process of Financial Accounting

The basic process is:

1. Recording → 2. Classifying → 3. Summarizing → 4. Reporting/Communicating

1. Recording

Recording/logging the day-to-day transactions of a business.

Example:

- ₹50,000 sales
- ₹10,000 rent expense
- ₹20,000 purchase

This historically developed into **bookkeeping and double-entry bookkeeping**.

2. Classifying

Grouping similar transactions under appropriate heads.

Example:

Transaction	Classification
Product sales	Sales
Rent	Expenses
Machinery purchase	Investment/Asset

3. Summarizing

Converting classified financial information into financial reports.

Important reports:

- **Profit & Loss Account / Income Statement**
- **Balance Sheet**
- **Cash Flow Statement**

4. Reporting & Communicating

Financial information is provided to users so they can:

- Understand the business
- Interpret information
- Analyze performance
- Make decisions

3. Why Do Businesses Need Accounting?

1. Record business activities

Business Data Management (Part - 3)

Accounting captures and records the company's **economic activities and transactions** meaningfully.

2. Measure company performance

It helps answer:

Is the company making money or losing money?

Management, shareholders and other users can track performance.

3. Create a common language

Companies need a **standardized language** for communicating their economic activities.

Accounting provides a common format for presenting financial information.

4. Build trust and transparency

Accounting follows established:

- Rules
- Guidelines
- Accounting principles
- Legal requirements

This makes financial information more **reliable, transparent and comparable**.

4. Types of Accounting

The lecture discusses **three broad types**:

Type	Main Focus
Financial Accounting	Recording, classifying, summarizing and reporting transactions
Cost &	Internal cost

Type	Main Focus
Management Accounting	control and decision-making
Tax Accounting	Tax-related rules, obligations and requirements

Financial Accounting

Focuses mainly on recording business transactions and reporting financial information to **internal and external users**.

Cost & Management Accounting

Primarily focuses on **internal organizational decisions**.

It helps management understand:

- Costs
- Expenses
- Cost control
- Efficiency
- Investment decisions

Example:

A company wants to decide whether investing in new machinery would benefit the organization.

Tax Accounting

Focuses specifically on **taxation** and compliance with tax rules and regulations prescribed by the relevant government/regulator.

5. Users of Accounting Information

There are two major categories:

A. Internal Users

Users **within the organization**.

Business Data Management (Part - 3)

Examples:

- Management
- Employees
- Different departments
- Internal data teams

Purpose

They use accounting information for:

- Managing the company
- Monitoring performance
- Making decisions
- Analyzing operations

B. External Users

Users **outside the organization** but connected to or interested in the company.

Examples:

- Investors
- Shareholders
- Lenders/Banks
- Suppliers
- Government agencies
- External analysts
- Credit analysts

Example

If a bank gives a company a loan, the bank may examine the company's financial information to understand **how the company is performing and whether it can meet its obligations.**

6. GAAP

GAAP = Generally Accepted Accounting Principles

GAAP is a set of **standard rules and guidelines** followed when preparing financial accounting information and financial statements.

Why is GAAP needed?

Imagine every company used completely different accounting rules.

It would become difficult to:

- Compare companies
- Analyze financial information
- Understand financial statements
- Trust the reported information

Therefore, GAAP provides **standardization and consistency.**

Indian context

The lecture mentions the **Accounting Standards Board**, which draws many principles from generally accepted accounting principles.

7. Three Important Accounting Principles

1. Revenue Recognition Principle Meaning

Revenue/sales should be recognized when the **goods or services are provided/accepted by the customer and the revenue is earned**, regardless of when cash is actually received.

Example

Business Data Management (Part - 3)

A company delivers headphones to a customer in **October**.

Customer pays in **December**.

Revenue is recognized in October, not December.

Remember:

Revenue recognition depends on when revenue is earned, not simply when cash is received.

8. Expense Recognition / Matching Principle

Meaning

An expense should be recognized in the **same period as the revenue that the expense helped generate.**

Example

A salesperson receives commission for generating sales.

If the sales happen in **March**, the related commission expense should be recognized with those sales, even if the payment occurs later.

Remember:

Match the expense with the revenue it helped generate.

9. Cost / Measurement Principle

Meaning

An asset is recorded at its **original historical cost**, rather than simply changing its recorded value

according to its current market value.

Example

A company buys machinery for: **₹5 lakh in 2010**

Even if its market value today is:

- ₹8 lakh, or
- ₹3 lakh

the machinery is recorded based on its **historical cost of ₹5 lakh**, according to the principle described in the lecture.

Remember:

Historical cost →

Original purchase price



Asset side of the Balance Sheet

1. Two Important Aspects of a Business

Every business has:

1. **Assets** → Things the business **owns or controls**.
2. **Liabilities** → Obligations/amounts the business **owes to outsiders**.

These are represented in the **Balance Sheet**.

Balance Sheet

A statement showing the financial position of a business **on a specific date**.

Basic relationship:

Total Assets = Total Liabilities + Owner's Equity

Business Data Management (Part - 3)

2. What is an Asset?

An **asset** is something that a business **owns or controls** and which is expected to provide **future economic benefit**.

Simple example

An Uber driver's car is an **asset** because the car helps the driver earn money in the future.

Basic conditions of an asset

Something can be considered an asset when:

1. It is **owned or controlled** by the company.
2. It generates or helps generate **future economic value/benefit**.
3. It may provide a **competitive advantage** to the company.

3. Types of Assets

Broadly, assets can be classified into:

A. Fixed Assets

B. Current Assets

C. Investments

D. Intangible Assets

4. Fixed Assets

Fixed assets are assets used by a business for a **long period**, generally **more than 12 months**, to help generate revenue.

Examples

- Land
- Buildings

- Plant
- Machinery
- Furniture
- Fixtures

Example

A manufacturing company uses a machine to produce products for several years.

Therefore:

Machine → Fixed Asset

Important point

The purpose for which something is used matters.

Suppose:

- An Uber driver owns a car → **Asset**
- A car dealer buys cars to sell them → **Product/Inventory**, not a fixed asset

So:

Used to generate

future revenue →

Asset

Bought for resale →

Product/Inventory

5. Current Assets

Current assets are assets that are generally **liquid or easily convertible into cash** and are used for the **day-to-day operations** of a business.

Examples

- Cash
- Bank balance
- Debtors/Receivables
- Inventory

Business Data Management (Part - 3)

- Short-term investments
- Loans and advances
- Prepaid expenses

Main idea

Current assets help a business run its daily operations.

6. Cash and Bank Balance

Businesses need cash and bank balances to:

- Buy goods
- Pay expenses
- Run daily operations
- Conduct business transactions

Therefore:

Cash + Bank Balance → Current Assets

7. Marketable Securities / Short-Term Investments

Companies may keep surplus cash in:

- Marketable securities
- Short-term investments

These can be converted into cash relatively easily and therefore form part of **current assets**.

8. Debtors / Receivables

Companies can sell goods in two ways:

- **Cash sale**
- **Credit sale**

When a company sells goods on credit, the customer has not yet paid the company.

Such customers are called:

Debtors / Receivables

Example

Company sells ₹50,000 worth of goods to Rahul on credit.

Rahul has not paid yet.

Therefore:

Rahul =

Debtor/Receivable

For the company:

Debtors → Current Asset

because the company expects to receive cash from them.

9. Inventory

Inventory refers to goods/materials held by a business for production or sale.

It may include:

1. Raw Materials

Materials used to manufacture products.

2. Work in Progress (WIP)

Products that are still being manufactured.

3. Finished Goods

Completed products ready for sale.

Therefore:

Raw Materials + WIP + Finished Goods = Inventory

And:

Business Data Management (Part - 3)

Inventory → Current Asset

10. Loans and Advances

Sometimes companies have surplus cash and provide:

- Loans
- Advances

to other organizations or third parties.

These can also be classified as **current assets** when they are expected to be recovered within the relevant short-term period.

11. Prepaid Expenses

Sometimes a company pays an expense **in advance**.

Example

A company pays its insurance for the entire year at the beginning of the year.

The company has already paid, but it will receive the insurance benefit over the year.

Such prepaid expenses are treated as:

Current Assets

because they represent a future benefit.

12. Main Current Assets to Remember

For exams, remember these important ones:

Current Asset	Meaning
---------------	---------

Current Asset	Meaning
Cash	Money held by the business
Bank Balance	Money kept in bank accounts
Debtors/Receivables	Customers who owe money
Inventory	Raw materials, WIP and finished goods
Short-term Investments	Investments intended for short-term purposes
Prepaid Expenses	Expenses paid in advance
Loans & Advances	Amounts given to others that are expected to be recovered

13. Long-Term Investments

Businesses may also invest surplus money for the **long term**.

Examples:

- Government securities
- Government bonds

Business Data Management (Part - 3)

- Equity investments
- Investments in other companies
- Investments in startups

These are shown separately from normal operating assets.

Strategic Investment

A company may invest in another company to:

- Expand its market
- Build a strategic partnership
- Gain business advantages

Example from the lecture:

Tata Motors holding a large stake in Jaguar Land Rover.

14. Tangible Assets

Tangible assets are assets that have a **physical form** and can be touched or felt.

Examples

- Land
- Building
- Machinery
- Plant
- Furniture

Easy memory:

Tangible = Touch

15. Intangible Assets

Intangible assets are assets that **do not have a physical form**, but can still provide economic value to a business.

Examples

- Patents

- Software
- Source code
- Apps
- Operating systems
- Platforms
- Customer databases

Easy memory:

Intangible = Cannot physically touch

Example: Patent

A pharmaceutical company develops a new drug and obtains a **patent**.

The patent itself cannot be physically touched, but it gives the company valuable rights.

Therefore:

Patent → Intangible Asset

16. Asset vs Product — Very Important

The same physical item can be an **asset for one business** but **inventory/product for another**, depending on its purpose.

Example: Car

Uber driver:

Car → used to provide transportation and earn money
→ **Asset**

Car dealer:

Car → bought to sell to customers
→ **Product/Inventory**

Remember:

Business Data Management (Part - 3)

**Purpose of use
determines the
classification.**



Liability side of the Balance Sheet

1. Liability

A **liability** is something that a business **owes to another party** and must pay or settle in the future.

Examples

- Money borrowed from a bank
- Money owed to suppliers
- Salaries owed to employees
- Taxes payable to the government

Simple example

A shop buys ₹2 lakh of phones **on credit**.

- Phones received today
- ₹2 lakh has to be paid later
- Therefore, ₹2 lakh is a **liability**

2. Characteristics of a Liability

For something to be considered a liability, generally these three conditions apply:

1. Past transaction

The liability arises because of something that has already happened.

Example: The business purchased goods on credit.

2. Future outflow of resources

The business will have to give something in the future, usually:

- Cash
- Goods
- Services

3. Obligation

The business has a responsibility to settle the amount.

Example: A bank loan creates an obligation to repay the loan plus interest.

Easy way to remember

**Past transaction → Future
payment → Present obligation**

3. Types of Liabilities

Liabilities are broadly divided into:

1. **Current Liabilities**
2. **Non-current Liabilities**

Type	Payment period	Main purpose
Current liability	Usually within 12 months	Day-to-day operations
Non-current liability	More than 12 months	Long-term financing/assets

4. Current Liabilities

Current liabilities are obligations that the business normally needs to settle **within 12 months**.

They generally arise from the **day-to-day operations** of the business.

Examples

Business Data Management (Part - 3)

- Creditors
- Outstanding expenses
- Tax payable
- Accounts payable
- Short-term loans

A. Creditors

A **creditor** is a person or business to whom the company owes money because it purchased goods or raw materials **on credit**.

Example

A shop buys ₹50,000 of goods from a supplier on credit.

The supplier is the **creditor**.

The shop has a liability of **₹50,000**.

B. Outstanding Expenses

An **outstanding expense** is an expense that has already been incurred but **has not yet been paid**.

Example

Employees have earned ₹30,000 salary, but the company has not paid them yet.

₹30,000 = **Outstanding salary** → **Current liability**

C. Tax Liability

If a company earns profit, it may have to pay taxes to the government.

Until the tax is paid, the amount payable is a **tax liability**.

D. Accounts Payable / Notes Payable

Accounts Payable

Money that the business owes to suppliers for goods or services purchased on credit.

Notes Payable

A formal written promise by the business to pay a certain amount to another party at a specified time.

Both can be classified as **current liabilities** when they are due within the short term.

5. Current Liability Example

An electronic shop purchases phones worth **₹2,00,000 on credit**. Payment must be made within **30 days**.

Analysis

- Past transaction? → **Yes**
- Future cash payment? → **Yes**
- Obligation to pay? → **Yes**
- Payment within 12 months? → **Yes**

Therefore:

₹2,00,000 = Current Liability

6. Non-Current Liabilities

Non-current liabilities are obligations that are normally payable **after more than 12 months**.

They are generally used to finance **long-term assets** or long-term business requirements.

Examples

- Long-term bank loans

Business Data Management (Part - 3)

- Long-term mortgages
- Loans for purchasing machinery
- Loans for purchasing buildings

Example

A company takes a **₹20 lakh loan** from a bank and agrees to repay it over **5 years**.

Therefore, it is a:

Non-current liability / Long-term liability

7. Long-Term Loans

A **long-term loan** is money borrowed by a business that will be repaid over a period of **more than one year**.

The business normally has to pay:

Principal + Interest

Example

Loan = ₹20,00,000

Period = 5 years

The company must repay the borrowed amount according to the loan agreement and also pay the required **interest**.

8. Why Take Long-Term Loans?

Long-term loans are generally used to finance **long-term assets**, such as:

- Land
- Buildings
- Machinery
- Plants
- Other major business assets

Basic idea

Long-term loan → Long-term financing → Long-term asset

9. Secured and Unsecured Loans

Loans can also be classified based on whether the lender receives **security/collateral**.

There are two types:

1. **Secured loan**
2. **Unsecured loan**

A. Secured Loan

A **secured loan** is a loan backed by an **asset used as security**.

The asset is called **collateral**.

Example

A business takes a ₹10 lakh loan and provides property as collateral. If the borrower fails to repay according to the agreement, the lender has rights over the collateral, subject to the loan terms and applicable law.

Common collateral

- Property
- Gold
- Machinery
- Other valuable assets

Advantage

Because the lender has security, the lender's risk is generally lower. Therefore, secured loans often have:

Lower interest rate

10. Unsecured Loan

Business Data Management (Part - 3)

An **unsecured loan** is a loan that is **not backed by specific collateral**.

The lender does not have a particular asset pledged as security for the loan.

Because the lender generally takes more risk, unsecured loans often have:

Higher interest rate

11. Secured vs Unsecured Loan

Feature	Secured Loan	Unsecured Loan
Collateral	Yes	No
Lender's risk	Generally lower	Generally higher
Interest rate	Usually lower	Usually higher
Example	Loan against property	Loan without specific collateral

Remember:

More security → Lower risk →

Usually lower interest

Less security → Higher risk →

Usually higher interest

12. Current vs Non-Current Liability

Current Liability	Non-Current Liability
Usually payable within 12 months	Payable after 12 months
Related mainly	Generally related

Current Liability	Non-Current Liability
to daily operations	to long-term financing
Creditors	Long-term bank loans
Outstanding salaries	Mortgage
Tax payable	Loan for machinery
Accounts payable	Long-term borrowing

13. Important Terms

Term	Meaning
Liability	Amount/obligation a business owes
Obligation	A responsibility to pay or provide something
Creditor	Person/business to whom money is owed
Outstanding expense	Expense incurred but not yet paid
Accounts payable	Amount owed to suppliers for credit purchases
Notes payable	Formal written promise to pay
Current liability	Liability generally due within 12 months
Non-current	Liability generally

Business Data Management (Part - 3)

Term	Meaning
liability	due after 12 months
Loan	Money borrowed that must be repaid
Interest	Cost paid for borrowing money
Secured loan	Loan backed by collateral
Unsecured loan	Loan without specific collateral
Collateral	Asset given as security for a loan
Lender	Person/institution that provides a loan
Borrower	Person/business that takes a loan
Principal	Original amount borrowed
Repay	To pay back borrowed money
Tax liability	Tax amount that the business has to pay
Long-term asset	Asset used for a long period, generally beyond one year
Financial obligation	Amount the business is required to pay



Capital and The Accounting Equation

1. Accounting Equation

The fundamental accounting equation is:

$$\underline{\text{Assets} = \text{Liabilities} + \text{Owner's Equity}}$$

- **Assets** → What the business owns.
- **Liabilities** → What the business owes to outsiders.
- **Owner's Equity** → Owner/shareholders' money invested in the business plus accumulated profits/reserves.

The equation must always balance.

2. Owner's Equity / Equity Capital Meaning

Equity capital is the **owner's/shareholders' money invested in the business.**

Over time, equity changes because of:

- Initial capital invested → **increases equity**
- Accumulated profits → **increase equity**
- Accumulated losses → **decrease equity**
- Owner withdrawals → **decrease equity**

Basic idea

$$\underline{\text{Equity} = \text{Capital} + \text{Reserves and Surpl} \\ \underline{\text{us}}}$$

3. Accumulated Profits

Accumulated profits = Profits kept in the business over time instead of being distributed or withdrawn.

Business Data Management (Part - 3)

These are generally included in **reserves**.

Examples of reserves mentioned:

- Profit & Loss balance
- General reserve
- Capital reserve
- Securities premium

4. Capital Reserve

A **capital reserve** arises from profits made from **non-operating activities**.

Example:

A company sells machinery for more than its relevant value/cost and makes a profit.

That profit can form part of the **capital reserve**.

5. Securities Premium

When a company issues shares for more than their **face value**, the extra amount is called **securities/share premium**.

Example:

Face value of share = ₹10

Issue price = ₹50

Securities Premium = $50 - 10 = ₹40$

6. Reserves & Surplus

Reserves and surplus can include:

- Profit & Loss balance
- General reserves
- Capital reserves
- Securities premium
- Other accumulated reserves

Therefore:

$\text{Equity} = \text{Initial Capital} + \text{Reserves \& Surplus}$

7. Equity Capital vs Preference Capital

Businesses can have different forms of share capital. Two important ones are:

1. **Equity share capital**
2. **Preference share capital**

Main Differences

Feature	Equity Shareholders	Preference Shareholders
Return	No fixed return	Fixed/guaranteed dividend
Profit potential	Higher	Limited to fixed return
Repayment during liquidation	After preference shareholders	Before equity shareholders
Voting rights	Have voting rights	Generally do not have voting rights
Risk	Higher	Lower

Why equity is attractive

Equity shareholders have **unlimited upside potential**.

If the company performs extremely well, equity shareholders can receive much higher returns than the fixed return of preference shareholders.

Business Data Management (Part - 3)

8. Preference in Preference Shares

Preference shareholders receive preference mainly in two areas:

1. Dividend

They receive a **fixed dividend** before equity shareholders.

Example:

Preference capital = ₹5 crore

Fixed dividend = 10%

₹5 crore $\times$ 10% = ₹50 lakh

2. Repayment during liquidation

If the company is wound up/liquidated, preference shareholders are repaid **before equity shareholders**.

Example:

Preference capital = ₹100

Equity capital = ₹100

Amount available during liquidation = ₹150

Preference shareholders $\rightarrow$ ₹100

Remaining for equity shareholders $\rightarrow$ ₹50

9. Accounting Equation — What It Means

Everything a company owns (**Assets**) has been financed through either:

- **Liabilities** $\rightarrow$ money owed to outsiders
- **Equity** $\rightarrow$ owner's/shareholders' money

Therefore:

Assets = Liabilities + Equity

10. Example 1 — Starting a Business

Alpha Consulting starts with ₹50,000 invested by the owner.

Assets

Cash = ₹50,000

Liabilities

₹0

Equity

₹50,000

Therefore:

$50,000 = 0 + 50,000$

✓ Equation balances.

11. Example 2 — Buying on Credit

Alpha Consulting buys computer servers worth ₹10,000 **on credit**.

The company has not paid cash yet.

Assets

Cash = ₹50,000

Servers = ₹10,000

Total Assets = ₹60,000

Liabilities

Amount owed to supplier = ₹10,000

Equity

₹50,000

Therefore:

$60,000 = 10,000 + 50,000$

✓ Equation balances.

Important point

Buying something **on credit**:

- Assets $\uparrow$
- Liabilities $\uparrow$
- Equity unchanged

Business Data Management (Part - 3)

12. Example 3 — Paying the Supplier

The company pays ₹4,000 to the supplier.

Before payment:

- Assets = ₹60,000
- Liability = ₹10,000
- Equity = ₹50,000

After paying ₹4,000:

Assets

$60,000 - 4,000 = ₹56,000$

Liability

$10,000 - 4,000 = ₹6,000$

Equity

₹50,000

Therefore:

$56,000 = 6,000 + 50,000$

✓ Equation still balances.

13. Quick Rules to Remember

Owner invests money

Assets ↑, Equity ↑

Business borrows money

Assets ↑, Liabilities ↑

Business buys something on credit

Assets ↑, Liabilities ↑

Business pays a liability

Assets ↓, Liabilities ↓

Owner withdraws money

Assets ↓, Equity ↓

★ Most Important for Exam

Remember these four:

Assets = Liabilities + Equity

Equity = Capital + Reserves & Surplus

Preference shareholders: fixed dividend + priority in repayment.

Equity shareholders: voting rights + potentially unlimited returns.



Business Accounting for Data

Analysts

1. Why should a Data Analyst understand Accounting?

A Data Analyst doesn't just **calculate numbers**. They must understand what those numbers **mean for the business**.

Understanding accounting helps a Data Analyst:

1. **Understand business health**
2. **Interpret data correctly**
3. **Propose better business decisions**

2. Understanding Business Health

The balance sheet has three key components:

Assets = Liabilities + Equity

- **Assets** → What the business owns/controls
- **Liabilities** → What the business owes
- **Equity** → Owners' investment/value in the business

Example

ABC Ltd:

- Assets = ₹100 crore
- Liabilities = ₹85 crore
- Equity = ₹15 crore

Business Data Management (Part - 3)

$$100=85+15$$

What can a Data Analyst understand?

The company is **highly leveraged** because:

Liabilities $\gg$ Equity

A relatively small loss of ₹10–15 crore could potentially wipe out most/all of the ₹15 crore equity. Therefore, the business may appear **financially fragile/risky**.

⚠ This is only an initial interpretation. A final conclusion requires additional financial data.

3. Accounting Helps Interpret Data Correctly

A single metric can be misleading if viewed alone.

Example

ABC Ltd shows:

- Sales $\uparrow$ **25%**
- Cash $\downarrow$
- Liabilities $\uparrow$
- Receivables $\uparrow$

At first glance:

Sales increased by 25%
→ Business is doing well. **X**

But accounting knowledge tells us to investigate **receivables and cash collections**.

If sales are increasing but customers aren't paying:
Sales Growth $\uparrow$
but

Cash $\downarrow$, Receivables $\uparrow$

then growth may be **credit-driven rather than cash-supported**.

Data Analyst's interpretation

The company is generating more sales, but it may be struggling to **collect the money from customers**.

Therefore:

Revenue growth $\neq$ necessarily healthy growth

4. Accounting Helps Make Business Decisions

A Data Analyst should move beyond:

Data $\rightarrow$ Analysis

towards:

Data $\rightarrow$ Insight $\rightarrow$ Action

Example

Suppose ₹20 crore of cash is sitting idle in the company.

A Data Analyst may identify that the company isn't using its assets effectively to generate returns.

Possible actions to discuss with management:

1. **Invest in expansion/projects**
2. **Pay outstanding liabilities**
3. **Return money to shareholders as dividends**

The correct decision depends on the broader business situation.

5. Three Major Benefits for a Data Analyst

Business Data Management (Part - 3)

Accounting knowledge helps you	Meaning
Understand business health	Assess financial position
Interpret data accurately	Understand relationships between financial metrics
Recommend decisions	Convert analysis into actionable suggestions

Simple flow

Accounting Knowledge



Better Data Interpretation



Better Business Insights



Better Decisions

★ Key Examples to Remember

Example 1 — High Debt

Assets = ₹100 Cr, Liabilities = ₹85 Cr, Equity = ₹15 Cr

→ Highly leveraged

→ Equity is vulnerable to losses

→ Potentially higher financial risk

Example 2 — Sales Increasing

Sales ↑ 25% + Cash ↓ +

Receivables ↑ + Liabilities ↑

→ Don't immediately conclude the business is healthy.

→ Investigate collections.

→ Growth may be **credit-driven**.

Example 3 — Idle Cash

₹20 Cr cash sitting idle

→ Assets aren't being used productively.

→ Analyst can suggest evaluating:

Investment / Debt repayment / Dividends



Understanding the Income Statement

1. Income Statement / Profit & Loss Account

The **Profit & Loss (P&L) Account / Income Statement** shows the business's:

- **Revenue (Sales)** → money generated
- **Expenses** → costs incurred to generate revenue
- Ultimately helps understand **profitability**

2. Revenue / Sales (Top Line)

Revenue = Sales of goods or services

Examples:

- Consulting company → consulting fees
- Software company → software subscriptions
- Product company → product sales

Formula

Revenue = Price per unit × Units sold

Business Data Management (Part - 3)

Example: Ravi's Lemonade

- Price = ₹10/glass
- Average sales = 60 glasses/day
- June = 30 days

$60 \times 30 = 1,800$ glasses

$1,800 \times ₹10 = ₹18,000$

So, **June revenue = ₹18,000.**

3. Expenses

Expenses = Costs incurred to run the business and generate revenue.

Examples:

- Raw materials
- Salaries
- Rent
- Marketing
- Electricity/utilities

Important idea

Higher revenue does NOT automatically mean higher profit.

$\text{Profit} = \text{Revenue} - \text{Expenses}$

Example

Company	Revenue	Expenses	Profit
A	₹1,00,000	₹50,000	₹50,000
B	₹1,50,000	₹1,20,000	₹30,000

Although **B has higher revenue**, **A has higher profit** because A controls its expenses better.

Data analyst's focus: Don't look only at revenue; analyze **costs and profitability** too.

4. Variable vs Fixed Expenses

Variable Expenses

Expenses that **change with production/sales.**

Examples for lemonade:

- Lemons
- Sugar
- Salt
- Cups

More lemonade sold → more raw materials required → **higher variable cost.**

Less sold → **lower variable cost.**

Fixed Expenses

Expenses that generally **do not change with production/sales in the short term.**

Example:

- Shop rent

Whether Ravi sells 10 or 1,000 glasses, the rent generally remains the same.

Quick Comparison

Variable Expense	Fixed Expense
Changes with production/sales	Generally remains constant in short term
Lemons	Rent
Sugar	Basic fixed salary/rent-type costs

Business Data Management (Part - 3)

Variable Expense	Fixed Expense
Cups	—

Important: Fixed costs can increase when the business scales significantly, e.g. opening another shop.

5. Asset vs Expense

This is an important distinction.

Asset

Something the business **owns or controls that helps generate future revenue/value**.

Examples:

- Machinery
- Car used for business
- Juice machine

Expense

A cost whose value is **used up in running the business / generating current revenue**.

Examples:

- Fuel
- Rent
- Lemon
- Sugar
- Electricity

Example

Ravi buys a **juice machine** → **Asset**

Ravi buys **lemons and sugar** →

Expenses

Easy way to remember

Asset → helps generate
future value/revenue

Expense → value is

**consumed in running
the business**

6. Cash Sales vs Credit Sales

Cash Sale

Customer pays **immediately**.

Example:

Customer buys ₹50 lemonade and pays ₹50 immediately.

→ **Cash Sale**

Credit Sale

Customer receives the goods/services now but **pays later**.

Example:

Customer buys ₹50 lemonade today and promises to pay tomorrow.

→ **Credit Sale**

Businesses may provide credit periods such as:

- 30 days
- 60 days
- etc.

7. Receivables / Debtors

When a business makes a **credit sale**, the amount it expects to receive from the customer is called a:

Receivable / Debtor

Example:

Ravi sells ₹500 worth of lemonade on credit.

Receivable=₹500

The customer now owes Ravi ₹500.

8. Bad Debt

Business Data Management (Part - 3)

Sometimes a customer **doesn't pay** the amount owed.

That unpaid amount becomes a **Bad Debt**.

Example

Ravi makes ₹200 credit sales.

One customer owes ₹50 but cannot pay.

Bad Debt=₹50

The ₹50 becomes a **loss/expense** for Ravi's business because the money is not expected to come back.

9. Credit Sales Are Not Necessarily Bad

Credit sales are common, especially in **B2B businesses**.

Advantages

- Can help attract more customers
- Can increase sales/business

Risk

- Customers may delay payment
- Receivables can become **bad debts**
- Cash flow may suffer

Therefore, businesses should **monitor receivables and collections**.

10. Why This Matters to a Data Analyst

A data analyst should **not judge business performance using revenue alone**.

For example:

Sales ↑ 25%

but:

- Cash ↓
- Receivables ↑
- Liabilities ↑

This may indicate that sales growth is heavily **credit-driven**, and the company is struggling to collect cash.

So an analyst should examine:

$Revenue + Expenses + Cash Flow + Receivables$

rather than looking at revenue alone.



Advanced Timing Concepts and Dividends

1. Purchases

Purchases = Goods/raw materials bought by a business for **processing or resale**.

Examples:

- Manufacturer → buys raw materials
- Trading business → buys goods for resale

Types of Purchases

Cash Purchase

→ Payment is made immediately.

Credit Purchase

→ Payment is made later to the supplier.

Credit purchases create a **Creditor / Payable**.

Business Data Management (Part - 3)

**Credit purchase →
Payable (Liability)**

The unpaid amount still represents an expense because it has been incurred.

2. Credit Expenses

An expense can be incurred even when **cash has not yet been paid**.

Examples:

- Electricity
- Gas
- Water
- Salary
- Rent

If the expense is incurred now but payment happens later → **Credit Expense**.

Important

No cash payment ≠ No expense

The expense is recognized because the business has already incurred the cost.

Example — Ravi's Lemonade

Item	Amount	Type
Lemons	₹200	Cash purchase
Sugar	₹100	Credit purchase
Rent	₹300	Credit expense
Total Expense	₹600	

Even though ₹400 is unpaid:
Total Expense=₹600

3. Prepaid Expense

Prepaid expense = Payment made in advance before the expense is actually used/incurred.

The **unused portion** is treated as an **Asset** because it provides future benefit.

Example

Ravi pays **₹2,100 rent for 7 days**.

Daily rent:

$$2100 = ₹300$$

After Day 1:

- Expense used = ₹300
- Remaining prepaid amount:

$$2100 - 300 = ₹1,800$$

So:

Day 1	Amount
Expense	₹300
Prepaid expense (Asset)	₹1,800

Why is ₹1,800 an asset?

Because Ravi has already paid for something that will provide **future benefit** during the next 6 days.

**Prepaid Expense →
Asset**

4. Outstanding Expense

Outstanding expense = Expense that has been incurred but has not yet been paid.

It is also called an **accrued expense** in the video's context.

Business Data Management (Part - 3)

Because the business still has to pay it:

Outstanding Expense

→ **Liability**

Example

Weekly rent = ₹2,100

Daily rent:

$2100/7 = ₹300$

Ravi doesn't pay rent for 3 days:

$300 \times 3 = ₹900$

Therefore:

- Expense incurred = ₹900
- Cash paid = ₹0
- Outstanding expense = ₹900
- Liability = ₹900

Easy comparison

Prepaid Expense	Outstanding Expense
Paid before use	Used/incurred before payment
Future benefit	Amount still payable
Asset	Liability

★ **Remember**

Prepaid → Asset

Outstanding → Liability

5. Net Profit

The video gives the basic relationship:

Net Profit = Sales - Total Expenses

Example:

Revenue = ₹500

Expenses = ₹400

$500 - 400 = ₹100$ profit

6. Dividends

Dividend = Portion of the company's profit distributed to shareholders/owners.

It is **NOT** an expense.

Example

Revenue = ₹500

Expenses = ₹400

Profit = ₹100

Company decides:

- ₹60 → retained in business
- ₹40 → distributed as dividend

Dividend = ₹40

Important distinction

Expenses → occur before calculating profit

Dividend → distributed after profit is generated

So:

Revenue - Expenses = Profit

Then part of the profit may be: Dividend

7. Why Companies Pay Dividends?

Dividends can:

- Reward shareholders for investing
- Provide shareholders with income
- Reflect the company's history/ability of distributing profits

According to the video, if there is **no profit**, dividends cannot be distributed.

Business Data Management (Part - 3)

One-line memory trick

- **Prepaid = paid early → Asset**
- **Outstanding = payment late → Liability**
- **Dividend = profit distributed → Not Expense**

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

Asset → Future benefit

Liability → Future payment

Equity → Owner's claim

Revenue → Money earned

Expense → Money consumed

Profit → Revenue - Expense

Cash sale → Money NOW

Credit sale → Money LATER

Receivable → Customer owes US

Payable → WE owe someone

Bad debt → Receivable not collected

Prepaid → Paid NOW, benefit LATER → ASSET

Outstanding → Expense NOW, pay LATER → LIABILITY

Dividend → Profit distributed
→ NOT an expense

Variable → Changes with volume

Fixed → Usually stays fixed short-term

The most important things to remember

1. The master formula

For the P&L:

$$\text{Revenue} - \text{Expenses} = \text{Profit}$$

And for the Balance Sheet:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

These two equations are the backbone.

Trick

Revenue → earns

Expense → consumes

Asset → future benefit

Liability → future payment

Receivable = Receive later

Payable = Pay later

Can't Receive → Bad Debt

PREPAID = PAY → USE → ASSET

OUTSTANDING = USE → PAY → LIABILITY

Future benefit = Asset

Consumed benefit = Expense